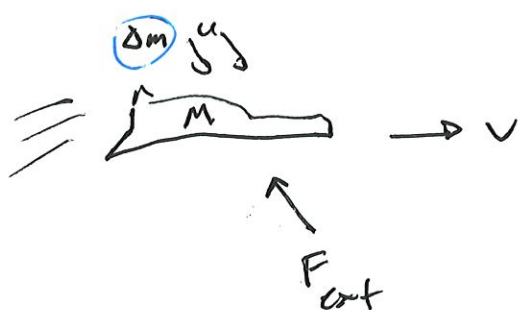


Impulse (review)

$$\vec{F} = \frac{d\vec{P}}{dt} \rightarrow \int_0^t \vec{F} dt = \vec{P}(t) - \vec{P}(0)$$

ex 4.12



Before

$$\vec{P}_i = M(t)\vec{v} + (\Delta m)\vec{u}$$

$$\vec{P}_i = \vec{P}(t)$$

$$\vec{P}_f = \vec{P}(t + \Delta t)$$

After

$$\vec{P}_f = (M + \Delta m)\vec{v}$$

but w/ calculus

→ this is a continuous process

$$\Delta \vec{P} = \vec{P}(t + \Delta t) - \vec{P}(t)$$

$$= (\vec{v} - \vec{u}) \Delta m$$

then

$$\frac{\Delta \vec{P}}{\Delta t} = (\vec{v} - \vec{u}) \frac{\Delta m}{\Delta t}$$

lim
 $\Delta t \rightarrow 0$

$$\frac{d\vec{P}}{dt} = (\vec{v} - \vec{u}) \frac{dm}{dt} = \vec{F}$$

Newton

therefore, the ext force is

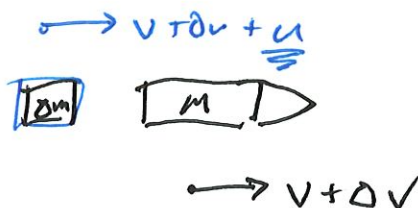
$$\boxed{\vec{F}_{ext} = (\vec{v} - \vec{u}) \frac{dm}{dt}}$$

Rocket Thrust (Ch 4.8)

time: t



time: $t + \Delta t$



$$\vec{P}(t) = (M + \Delta m) \vec{v}$$

$$\vec{P}(t + \Delta t) = M(\vec{v} + \Delta \vec{v}) + \Delta m(\vec{v} + \Delta \vec{v} + \vec{u})$$

$$\therefore \Delta \vec{P} = M \Delta \vec{v} + \Delta m(\Delta \vec{v} + \vec{u})$$

$$\frac{d\vec{P}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{P}}{\Delta t} = M \frac{d\vec{v}}{dt} + \frac{dm}{dt} (\cancel{0} \vec{v} + \vec{u})$$

$\cancel{0}$, this is 2nd order small

no ext forces!

$$\underbrace{\Sigma \vec{F}_{\text{ext}} = 0}_0 = M \frac{d\vec{v}}{dt} + \frac{dm}{dt} \vec{u}$$

$$0 = M \frac{d\vec{v}}{dt} - \frac{dM}{dt} \vec{u}$$

But what about $\Delta m \frac{d\vec{v}}{dt}$?

$$\dot{m} = -\dot{M}$$

Because $\frac{d}{dt}(m + M) = 0$

for take off

$$Mg = M \frac{d\vec{v}}{dt} - \frac{dM}{dt} \vec{u}$$

a small #, see Saturn V rocket in Ex 4.18!

4.16 Ex: Rocket in free space

$$M \frac{d\vec{v}}{dt} - \vec{u} \frac{dM}{dt} = 0$$

→ let's compute velocity

$$\frac{d\vec{v}}{dt} = \vec{u} \frac{1}{M} \frac{dM}{dt}$$

→ Integrate both sides

$$\int_{\vec{v}_0}^{\vec{v}_f} \frac{d\vec{v}}{dt} dt = \vec{u} \int_{t_0}^{t_f} \frac{1}{M} \frac{dM}{dt} dt$$

$$\vec{v}_f - \vec{v}_0 = \vec{u} \int_{M_0}^{M_f} \frac{dM}{M}$$

$$= \vec{u} \ln \frac{M_f}{M_0}$$

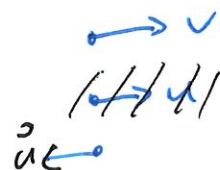
$$= -\vec{u} \ln \frac{M_0}{M_f}$$

$M_f < M_0$
so $\frac{M_f}{M_0} < 1$

$$\vec{v}_f = \vec{v}_0 - \vec{u} \ln \frac{M_0}{M_f}$$

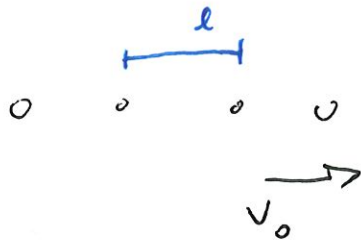
send fuel backwards

if you want to move forward



→ $-\vec{u}$
 $\vec{v}_f$
is forward (!)

Momentum Flow



quantized droplets

like a paddle

Impulse

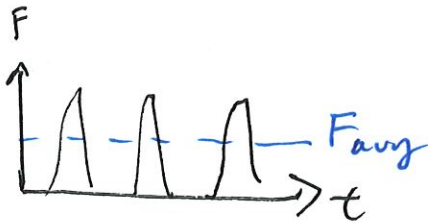
$$I = \int_{\text{on droplet}} F dx$$

$$= 0p$$

$$= m(v_f - v)$$

$$= -mv$$

assume 0
the droplet does not move



then

$$I_{\text{on hand}} = -I_{\text{on droplet}} = mv$$

If the spacing is l ,

then the period between impulses is

$$T = \frac{l}{v_0}$$

so the average force is

$$F_{\text{avg}} = \frac{I}{T} = \frac{mv}{l/v_0} = \frac{mv^2}{l}$$

Consider a column of moving fluid

if $\rho = m/v$

A $\vec{v}$

linear density = ρA

momentum/length = $\rho A \vec{v}$

$$\vec{P} = \text{Momentum flow} = \left(\frac{\text{momentum}}{\text{length}} \right) \cdot (\text{velocity}) = (\rho A v^2) \hat{v}$$

reflection

$$\rho A \vec{v} = \left[\frac{m}{v} A \frac{L}{T} \right] = \left[\frac{m}{T} \right] \quad \text{"momentum per length"}$$

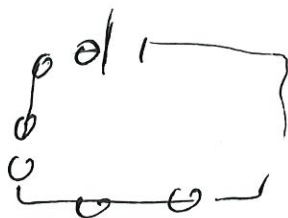
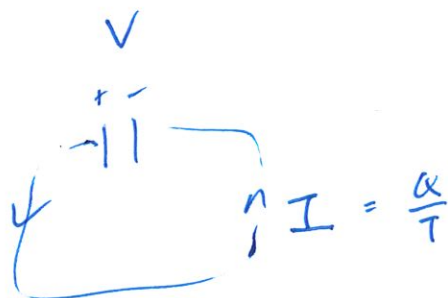
$$(\rho A \vec{v}) \Delta l = \left[\frac{M L}{T} \right] = [mv] \quad \text{regular momentum}$$

$$\text{Rate of Momentum Flow} = \dot{\vec{P}} = \rho A v \vec{v} = \left[\frac{mv}{T} \right] \\ \downarrow \\ \text{Momentum Flux}$$

$$\vec{J} = \rho v^2 \hat{v} = \frac{\dot{\vec{P}}}{A} = \left[\frac{mv/T}{A} \right] = \text{momentum flux density}$$

"charge flux" $\rightarrow$

charge: Q	Momentum: $p = mv$
current: $I = \frac{Q}{T} = \dot{Q}$	Momentum flux: $\dot{\vec{P}}$
current density: $\vec{J} = \frac{I}{A}$	Momentum flux density: $\vec{J} = \frac{\dot{\vec{P}}}{A}$



Q	I	$\vec{J}$
p	$\dot{\vec{P}}$	"momentum flux" (density)

current density: $\vec{J} = \frac{I}{A}$